

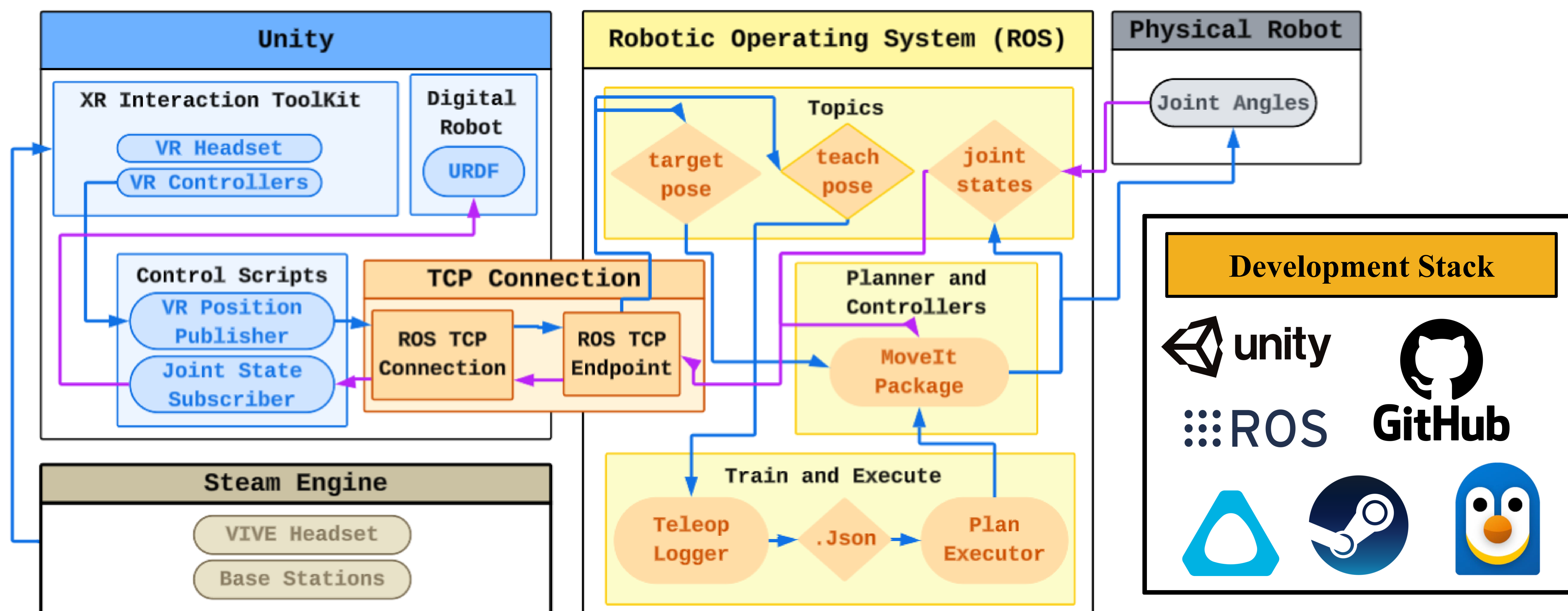
Abstract

For advanced robots to provide agency in autonomous environments, there is growing demand for robotic simulation and training to ensure safety, accelerate development, and reduce costs. This work proposes a rapid development pipeline for safe, efficient, and repeatable autonomous robotic systems. By integrating the Unity game engine, Robot Operating System (ROS), SteamVR, and a Vive headset, we enable digital twins, real-time teleoperation, environment simulation, and task planning. We demonstrate the system on a 6-DOF robotic arm, achieving intuitive teleoperation, ROS-based pose tracking, and autonomous replay of recorded trajectories.

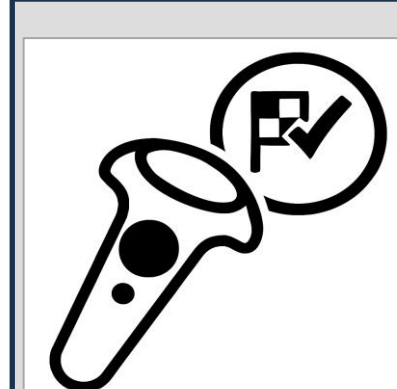
Impact and Applications

- ★ **Data Collection Made Simple**
 - Perform Task and Log Positions
 - ★ **Training Data**
 - Train Models on Human Control
 - ★ **Safe, Simulated Environments**
 - Test systems virtually, with no risk
 - ★ **High-Precision Human Control**
 - Medical, Construction, Assembly, etc..
 - ★ **Rapid Task Development**
 - Train, Record, Tune, Deploy
 - ★ **Remote Distributed Operation**
 - Robotic Control From Anywhere

System Architecture



Methods



Home Position and Orientation
- 10 seconds

Position Equation:

$$\text{final_pos} = \text{EE}_{\text{home}} + (\text{controller}_{\text{now}} - \text{controller}_{\text{home}})$$

Orientation Equation:

$$\text{final_rot} = \text{rot}_{\text{current}} \cdot \text{rot}_{\text{home}}^{-1}$$



Right Trigger:
Open Gripper by 25%



Left Trigger:
Close Gripper by 25%



Menu Button:
Log Current Position

System Capabilities

- ## ✓ Digital Twin Simulation



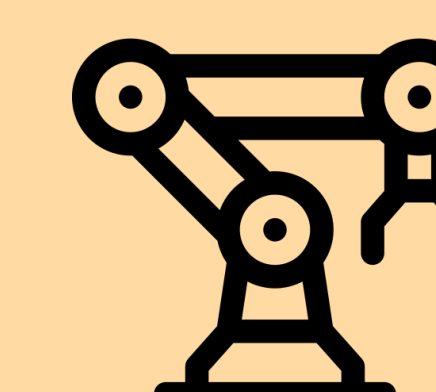
- ✓ Human-Robot Teleoperation



- ✓ Task Training



- ## ✓ Robot Task Automation



- ## ✓ Replayable Trajectories

Conclusion

- Accurate, real-time **Digital twin**
- Delayed **Robot Teleoperation**
- **Effective Task Training**
- **Repeatable Task**

Discussion

- Wide Range of Applications
- Performance limited by Robot and Processing specs
- Safe and Rapid Development for Robotics Development

References

NXROBO. *Sagittarius_ws*. GitHub repository.
https://github.com/NXROBO/sagittarius_ws